

우주비행학 (Space Flight Dynamics)

2018 년 봄학기

Home Work 3

Due date: April 17,

- 1) Problem 2 in Chapter 5
- 2) Problem 5 in Chapter 5
- 3) Problem 6 in Chapter 5
- 4) Problem 7 in Chapter 5
- 5) Problem 8 in Chapter 5

* The number of problem is referred by 2nd edition of the test book.

Four thrusters are mounted on a satellite with moment of inertia I , a distance R from the center of mass, as shown in Figure 5.21. The thrusters fire whenever the satellite strays from within an angular window (the “deadband”) of width θ , a technique referred to as “bang-bang” control. Argue that the satellite usually has angular momentum

$$|\mathbf{H}| \approx 2pR \quad (5.90)$$

and an average angular velocity

$$|\omega| \approx \frac{2pR}{I} \quad (5.91)$$

This causes the time interval between thruster firings to be about

$$\Delta t \approx \frac{I\theta}{2pR} \quad (5.92)$$

so the satellite eats through its fuel supply at a rate proportional to

$$\frac{p^2 R}{I\theta} \quad (5.93)$$

Why are small minimum-impulse-bit (MIB) thrusters desirable?

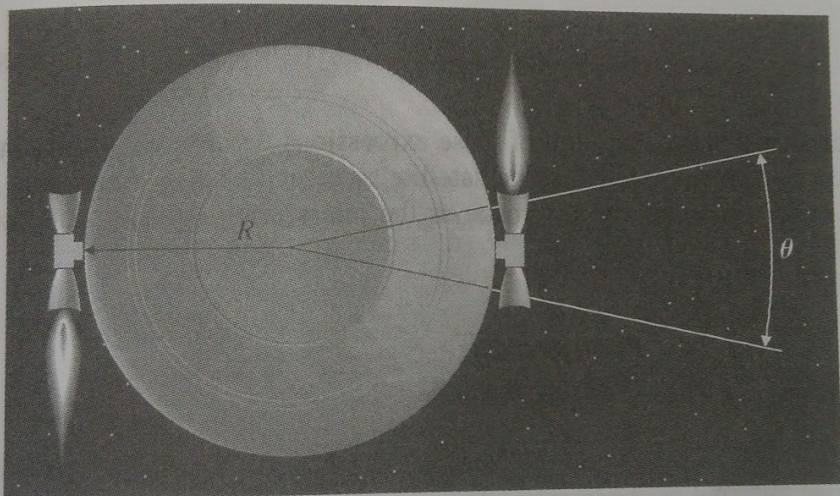


FIGURE 5.21

“Bang-bang” attitude control.

2. A spacecraft must sometimes be rotated about the intermediate inertia axis, although this is unstable. Write Euler's equations for a torque-free rigid body. For initial conditions $\omega_2 = \omega_{20}$ (not small) and ω_1, ω_3 (very small), show that Euler's equations imply $\omega_2 \approx \omega_{20}$, a constant, and

$$\begin{Bmatrix} A & 0 \\ 0 & C \end{Bmatrix} \begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_3 \end{pmatrix} + \begin{Bmatrix} 0 & (C-B)\omega_{20} \\ (B-A)\omega_{20} & 0 \end{Bmatrix} \begin{pmatrix} \omega_1 \\ \omega_3 \end{pmatrix} = 0 \quad (5.94)$$

These are a pair of linear, constant-coefficient equations. Substitute an assumed solution of the form

$$\begin{pmatrix} \omega_1 \\ \omega_3 \end{pmatrix} = \mathbf{A}e^{\lambda t} \quad (5.95)$$

and show that, as expected, rotation about the intermediate inertia axis is unstable.

Now, assume that the spacecraft has rate gyros, capable of supplying measured values of the spacecraft rates ω_1, ω_3 . Use these measured rates to produce control torques according to the feedback laws

$$M_1 = k\omega_3, \quad M_3 = k\omega_1 \quad (5.96)$$

where k is a gain constant. Show that the augmented system is stable when rotating about the intermediate axis if k/ω_{20} is between $|C - B|$ and $|B - A|$.

3. A dual-spin satellite shown in Figure 5.22, sometimes referred to as a *gyrostat*, has principal moments of inertia of A, B , and C about the axes $\mathbf{b}_1, \mathbf{b}_2$, and $\mathbf{b}_3$. It contains a rotor with moment of inertia I_r which spins on frictionless bearings with constant angular rate ω_r . If the angular-velocity vector of the satellite is $\boldsymbol{\omega}^{\text{bi}} = \{\omega_1, \omega_2, \omega_3\}^T$, show that the total angular momentum is

$$H^2 = A^2\omega_1^2 + B^2\omega_2^2 + (C\omega_3 + I_r\omega_r)^2 \quad (5.97)$$

while the total kinetic energy is

$$T = \frac{1}{2} (A\omega_1^2 + B\omega_2^2 + C\omega_3^2 + I_r\omega_r^2) \quad (5.98)$$

and that both are constant. Show that both of these expressions describe ellipsoids, but that one ellipsoid is no longer centered at the satellite's center of mass. Argue that it is still possible to define polhodes for a gyrostat as the intersection curves of the two ellipsoids.

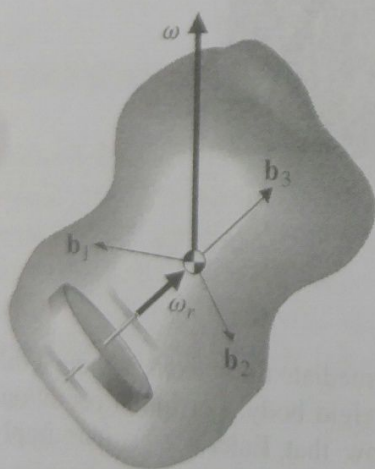


FIGURE 5.22
Dual-spin satellite.

4. A dual-spin satellite, as in the previous problem, is symmetric ($A = B$) and normally rotates about the $\mathbf{b}_3$ axis at angular rate

$$\omega_{30} = -\frac{I_r\omega_r}{C} \quad (5.99)$$

and thus has zero net angular momentum if ω_1 and ω_2 are zero. Write $\omega_3 = \omega_{30} + \delta\omega_3$, and write the expressions for the kinetic energy and angular-momentum ellipsoids. Argue that such a satellite can be controlled as if it were not spinning, although it does rotate.

5. The space-shuttle orbiter has a moment-of-inertia matrix

$$I = \begin{Bmatrix} 1.29 & 0 & 0 \\ 0 & 9.68 & 0 \\ 0 & 0 & 10.1 \end{Bmatrix} \times 10^6 \text{ kg m}^2 \quad (5.100)$$

in the principal-axis frame shown in Figure 5.23. Describe the stable gravity-gradient attitudes for the shuttle, and calculate the three oscillation frequencies for this vehicle in a 90-minute period orbit.

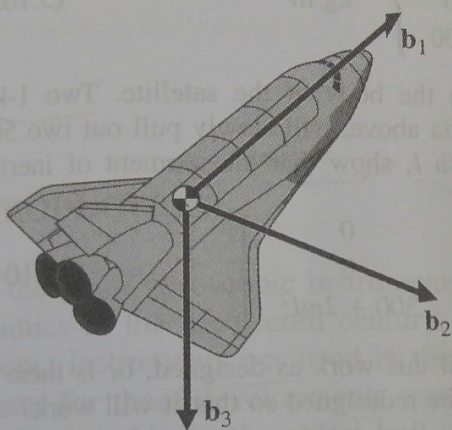


FIGURE 5.23

Space-shuttle principal axes.

6. Show that a single-mass yo-yo mechanism needs to have a string length of

$$l_f = \sqrt{R^2 + \frac{C}{m}} \quad (5.101)$$

to completely despin a satellite. However, unlike the two-mass yo-yo device, the satellite will not remain in its previous orbit. Calculate the velocity of the mass at release, and if the mass of the satellite is M , find the speed of the satellite with respect to the center of mass of the system after release.

7. To eliminate the rigors of the seasons, the government is studying a proposal to reorient the earth's spin axis normal to its orbital plane, eliminating the 23° tilt of the planet's axis shown in Figure 5.24. After the first attitude maneuver, how long will it take to precess the earth's axis to its new position? For the earth, $\dot{\psi} = 2\pi$ rad/day, and

$$\frac{A - C}{C} = 0.003279 \quad (5.102)$$

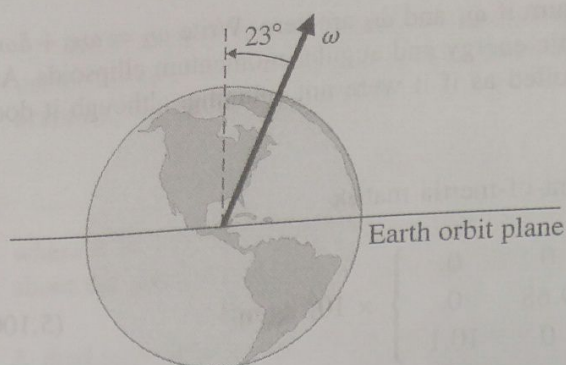


FIGURE 5.24

Reorientation maneuver for earth.

8. A satellite in Figure 5.25 initially rotates at $\omega^{bi} = 0.2\mathbf{b}_3$ rad/s, and has a moment-of-inertia tensor

$$I = \begin{Bmatrix} 500 & 0 & 0 \\ 0 & 600 & 0 \\ 0 & 0 & 300 \end{Bmatrix} \text{ kg m}^2 \quad (5.103)$$

Internal energy dissipation is negligible within the body of the satellite. Two 1-kg masses m , not included in the moments of inertia above, will slowly pull out two 50-m wire antennas. When the wires have a length l , show that the moment of inertia becomes

$$I_{\text{tot}} = \begin{Bmatrix} 500 & 0 & 0 \\ 0 & 600 + 2ml^2 & 0 \\ 0 & 0 & 300 + 2ml^2 \end{Bmatrix} \quad (5.104)$$

What is the final spin rate when $l = 50$ m? Will this work as designed, or is there a fatal flaw in this scheme? If so, can the scheme be redesigned so that it will work?

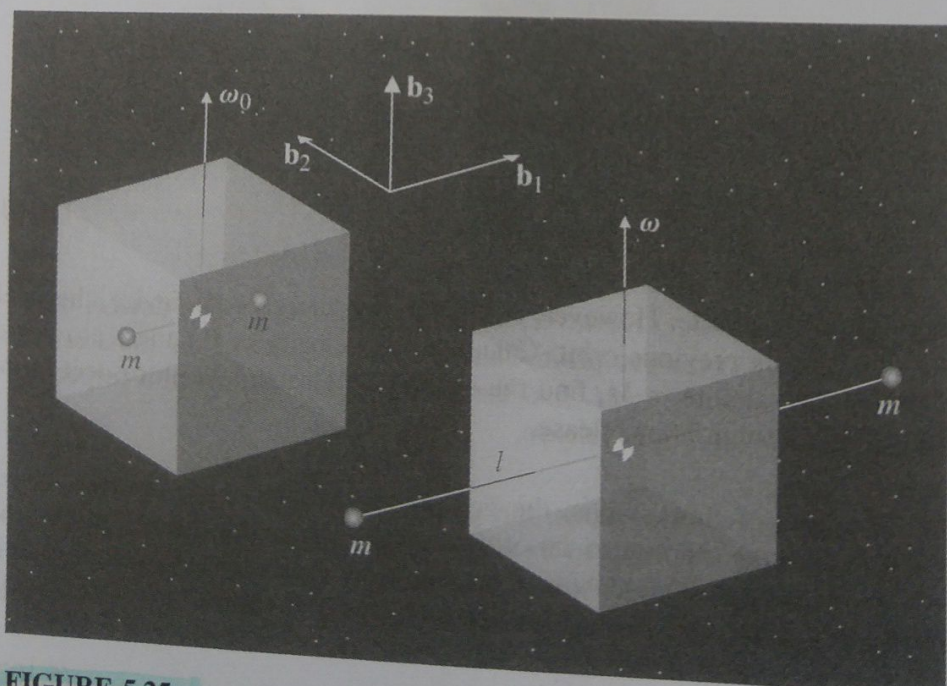


FIGURE 5.25

Antenna deployment from a spinning spacecraft.